Hybrid Classical-Quantum Autoencoder with Attention Mechanism for Time Series Anomaly Detection

Fatima Farhan
Department of Computer Science
National University of Computing and Emerging Sciences
Karachi, Pakistan
fatima.farhan2047@gmail.com

Lamia Asif
Department of Computer Science
National University of Computing and Emerging Sciences
Karachi, Pakistan
lamiaasif@gmail.com

Najaah Noor
Department of Computer Science
National University of Computing and Emerging Sciences
Karachi, Pakistan
ray.gitit@gmail.com

***Abstract*—Quantum computing offers potential advantages in pattern recognition and data compression for machine learning tasks. This paper proposes an extension to the quantum autoencoder (QAE) framework introduced by Frehner and Stockinger for unsupervised time series anomaly detection. We integrate three architectural enhancements: (a) a classical temporal attention network for intelligent feature selection prior to quantum encoding; (b) dual-axis quantum encoding combining RY and RZ rotations with IsingZZ inter-qubit interactions; and (c) an alternating CZ/CNOT entanglement pattern that accelerates quantum information mixing. The anomaly detection mechanism employs an ensemble score combining SWAP-test reconstruction fidelity with latent space entropy, enabling detection of both pattern deviations and model confidence collapse. Experiments on a synthetic ECG-like benchmark yield AUC-ROC of 0.8192 and F1 of 0.8095 with only 72 quantum trainable parameters. These results demonstrate that targeted architectural enhancements to quantum circuits can significantly improve anomaly detection capability without substantially increasing quantum resource requirements.**

***Index Terms*—quantum autoencoder, variational quantum circuit, anomaly detection, time series, temporal attention, hybrid quantum-classical learning**

I. Introduction

Time series anomaly detection is critical across electrocardiogram (ECG) analysis, industrial equipment monitoring, and IoT sensor networks [7]. The core challenge is asymmetric: normal data is abundant while anomalies are rare and largely unlabeled, making supervised approaches impractical. Unsupervised autoencoder methods address this by training exclusively on normal data and detecting anomalies via elevated reconstruction error [4]. LSTM-based encoder-decoders [6], convolutional recurrent autoencoders [8], and temporal convolutional variants [4] have demonstrated strong performance on sequential data, but share a fundamental limitation: large parameter counts — often tens of thousands — and substantial training data requirements that become prohibitive in resource-constrained or data-scarce settings. Variational hybrid classical-quantum architectures [3] offer an alternative, with parameterized quantum circuits (PQCs) learning compact data representations through superposition and entanglement at a fraction of the parameter cost of classical networks [2]. Frehner and Stockinger [1] applied this to time series anomaly detection, encoding windows into quantum states and training via the SWAP test as a compression loss. Validated on real ECG data and IBM quantum hardware, their quantum autoencoder (QAE) matched classical LSTM detection performance with 60–230× fewer parameters. Tscharke et al. [5] subsequently extended QAEs to multivariate settings, further establishing their practical viability. However, the base formulation of [1] leaves several opportunities unexplored: uniform fixed-index downsampling is agnostic to where anomaly-relevant information resides; single-axis RY encoding discards phase and inter-step correlation information; the Hardware-Efficient Ansatz (HEA) with linear CNOT topology has O(n) information mixing time; and single-signal detection based solely on SWAP-test fidelity is blind to anomalies with unusual latent representations. This paper proposes the Hybrid Attention-QAE, which addresses each limitation through targeted enhancements while preserving full compatibility with the training objective of [1]. The main contributions of this paper are: (1) A temporal attention preprocessing module that learns which time steps carry the most anomaly-discriminative information, replacing the fixed downsampling of [1] with a data-driven selection strategy. (2) Dual-axis quantum encoding combining RY, RZ, and IsingZZ gates, capturing amplitude, phase, and pairwise temporal correlations inaccessible to the single-axis scheme of [1]. (3) An alternating CZ/CNOT variational ansatz achieving O(log n) quantum information mixing, compared to O(n) in the linear topology of [1]. (4) An ensemble anomaly score combining SWAP-test reconstruction fidelity (0.6) and latent qubit entropy (0.4), providing dual-signal detection more robust than either signal alone. All experiments are implemented in PennyLane and PyTorch on Google Colab and evaluated on an identical balanced test subset across all three compared methods.

II. Literature Review and Related Work

A. Quantum Machine Learning Fundamentals

Quantum machine learning makes use of quantum mechanical properties to enhance classical algorithms. VQCs (variational quantum circuits) have emerged as a practical approach for NISQ (Noisy Intermediate-Scale Quantum) devices, where parametrized quantum gates are optimized through classical optimization loops.Recent advances in quantum machine learning have shown that such circuits can achieve strong parameter efficiency compared to classical approaches [1] [5]. This makes them particularly suitable for resource constrained applications in anomaly detection.

B. Quantum Autoencoders for Anomaly Detection

Frehner and Stockinger [1] introduced quantum autoencoders for ECG anomaly detection, demonstrating that quantum circuits can achieve competitive anomaly detection with extremely fewer parameters than classical alternatives. Their hardware efficient ansatz (HEA) utilized two-layer RY-RZ rotations with linear CNOT chains for entanglement. While this approach showed promise, the uniform encoding and fixed entanglement may not fully exploit quantum advantages for temporal pattern recognition [5]. Hybrid approaches, combining classical preprocessing and quantum circuits, have shown promising results to overcome these limitations [2][3]. These methods use classical networks for intelligent feature selection and quantum compression for nonlinear pattern learning [2]. Continued work in quantum autoencoder design has emphasized the importance of flexible ansatz design [5]. Observations suggest that deeper circuits with diverse entanglement patterns improve expressivity while maintaining feasibility on near-term quantum hardware.

C. Attention Mechanisms and Feature Selection in Time Series

Classical machine learning has benefited from learned feature selection [4] [7]. For time series anomaly detection tasks specifically, approaches that learn feature importance consistently outperform fixed sampling strategies, particularly in datasets with irregular anomaly locations [4]. This is very important in the case of time series where the anomalies are at unpredictable temporal positions. In a quantum setting, attention mechanisms are a natural extension of this principle when combined with quantum circuits [3].

D. Anomaly Detection Paradigms

Standard approaches treat anomaly detection as reconstruction: normal patterns compress well; abnormal data fails reconstruction [6]. However, this single-signal strategy can miss anomalies where reconstruction partially succeeds but model confidence collapses. Entropy-based approaches in classical settings complement reconstruction-focused methods [7]. In particular, the combination of reconstruction based signals and uncertainty measures on the latent representations can lead to a more robust anomaly detection [7]. Recent deep learning methods for time series anomaly detection use ensemble approaches that take advantage of multiple complementary signals [7] [8], a principle we apply to quantum autoencoders. Recent work extends these principles to quantum and hybrid architectures.

E. Hybrid Classical-Quantum Architectures

Recent advances stress hybrid designs where classical preprocessing or postprocessing augments quantum circuits [2] [3]. Classical networks often perform better at feature extraction while quantum circuits provide compression [2]. Attention-quantum integration follows this paradigm [3], using classical learning for intelligent downsampling and quantum processing for nonlinear pattern compression.

III. Methodology

A. Problem Formulation

For a time-series dataset $\{X, y\}$ where $X \in \mathbb{R}^{N\times L}$ represents N samples of length L and $y \in \{0, 1\}$ denotes normal vs anomalous labels, the objective is to learn a model that achieves high anomaly detection accuracy with minimal parameters. Following the base paper [1], the compression stage is limited only to 8 quantum qubits.

B. Attention-Based Feature Selection

Classical preprocessing begins with a small trainable neural network [2] that learns temporal importance weights from the input signal. A sequence of 140 points is given as an input, the attention network assigns a relative importance score to each time step. This highlights indices that contain the most relevant anomalous information. Based on these 140 scores, the 8 most important values are selected and weighted for quantum processing. This approach replaces the base paper's fixed interval downsampling strategy at regular intervals, an enhancement shown to be effective in hybrid quantum-classical architectures [3].

This design choice is motivated by observing the nature of ECG anomalies. Anomalies rarely occur uniformly and tend to appear at specific points in the signal, often forming localized patterns in time. By learning the regions containing most information, the attention mechanism can focus only on the relevant parts of the signal without depending on manual tuning. This adaptive approach to temporal feature extraction is particularly valuable in anomaly detection, where anomaly locations are inherently unpredictable [4].

C. Dual Quantum Encoding with Correlations

Each of the 8 attention-selected values is encoded into a corresponding qubit through three operations:

RY Rotation: The primary amplitude information is encoded using RY(πx_q), where x_q is the normalized value. This step is inherited from the base paper [1].

RZ Rotation: For capturing phase information, the RZ($2\pi x_q$) gate is applied. The 2π scaling allows a larger angle range. Together with RY, these two single-qubit rotations provide richer state initialization compared to using RY gate alone. This dual-axis encoding strategy [5] captures more complete quantum state information, improving the model's representational capacity.

Ising ZZ Interaction: For adjacent qubits q and $q+1$, IsingZZ gates are applied with angle $\pi x_q \times x_{q+1}$. When consecutive values are similar, the gate creates strong entanglement. When the values are dissimilar, entanglement is weak. These temporal correlations are directly encoded at the input stage rather than relying on the ansatz to learn them. Encoding data-dependent interactions at the input, as demonstrated in recent quantum autoencoder work [5], enables more efficient pattern recognition in time series data.

D. Alternating Variational Ansatz

The parametrized quantum circuit [3] consists of 3 layers. Each layer contains rotations and entanglement, alternating between two patterns to enable diverse quantum correlations:

Even Layers (0, 2):

- Each qubit undergoes universal rotations: RX(θ), RY(θ), RZ(θ).
- Entanglement is implemented by CZ gates in a checkerboard pattern: (0,1), (2,3), (4,5), (6,7) are connected in the first pass, then (1,2), (3,4), (5,6) in the second pass.
- This setup allows parallel information exchange between non-overlapping qubit pairs.

Odd Layers (1):

- Each qubit undergoes universal rotations: RX(θ), RY(θ), RZ(θ).
- Entanglement is implemented by a CNOT chain that follows 0→1→2→...→7 then wraps around from 7→0.
- This setup allows sequential information propagation.

Rationale for Alternation: In the fixed linear CNOT design of the base paper, qubit-7 needs 7 hops to send information to qubit-0. However, with the alternating pattern design, information reaches all qubits exponentially faster. Even layers create several parallel pathways, while the odd layers ensure that information flows through the entire circuit. This hybrid topology design takes inspiration from expander graphs in classical circuit design [4], where for fast mixing both parallel and sequential connections are required.

This 3-layer structure results in 72 learnable parameters (8 qubits $\times$ 3 rotations $\times$ 3 layers), compared to the base paper having 32 learnable parameters. This boosts expressivity while remaining approximately 735 times more efficient than classical LSTM-AE [6]. The increased expressivity is critical for capturing the diverse patterns present in anomalous time series data, as demonstrated in recent comparative studies.

E. Quantum State Measurement and Compression

After processing, the quantum circuit operates on 13 wires total:

- 8 input qubits that encode time series values
- 4 latent qubits for compressed representation
- 4 auxiliary qubits for SWAP test
- 1 control qubit for measurement

The 4 latent qubits store the learned compressed representation. The other 4 auxiliary qubits are discarded, which results in a 50% compression ratio. This compression mechanism is fundamental to quantum autoencoder-based anomaly detection [5], enabling efficient latent representation while maintaining discriminative power. A SWAP test [5] is performed using the control qubit to measure how closely the reconstructed output matches the input. This measurement principle, established in quantum information theory, provides the fidelity score essential for anomaly detection.

F. Ensemble Anomaly Scoring

Signal 1 — Reconstruction Fidelity (60% weight) [6]: The SWAP test produces a Z-basis measurement $z \in [-1, +1]$. This is converted to a fidelity score ($r = 1 - (1 + z)/2$), where $r \in [0, 1]$. Lower values indicate good reconstruction and normal data, while higher values show poor reconstruction and potential anomaly.

Signal 2 — Latent Space Entropy (40% weight - Novel Contribution) [7]: The 4 latent qubits are measured in the Z-basis, producing $\{z_0, z_1, z_2, z_3\}$. Each measurement is then converted to a probability using $p_q = (1 - z_q)/2$.

The binary Shannon entropy for each qubit is computed as:

$$h_q = -(p_q \log p_q + (1 - p_q) \log(1 - p_q)) \tag{1}$$

The average entropy across all 4 latent qubits is:

$$\text{ent} = \frac{1}{4} \sum h_q \tag{2}$$

Interpretation: Low entropy means qubits are in definite $|0\rangle$ or $|1\rangle$ states. This suggests that the model is confident in its compression. On the other hand, high entropy indicates qubits are in a state of superposition, reflecting uncertainty in latent representations and in the model itself. In many cases anomalous data lead to higher entropy because the learned pattern of normal data fails to hold. This principle aligns with established anomaly detection literature where model uncertainty is an important signal [7].

Final Score:

$$\text{score} = 0.6 \times r + 0.4 \times \text{ent} \tag{3}$$

The weights (60/40) give priority to the reconstruction error as the main signal while entropy is used as a complementary detector for model confusion. Ensemble approaches combining multiple anomaly signals have been shown effective in improving detection robustness [8]. The decision threshold is selected using Youden's J statistic on a validation set.

G. Training Procedure

The quantum circuit is trained with gradient-based optimization [2], using the Adam optimizer and a step size of 0.08. The objective is to minimize SWAP-test reconstruction loss over 35 epochs on 150 training samples. This reduced dataset size is chosen due to the high cost of simulation. Full quantum hardware deployment could scale to larger datasets. The classical attention network is trained separately to maximize feature variance on the normal training distribution.

IV. Results

A. Experimental Setup

Three models were evaluated on a synthetic ECG-like dataset comprising 3,000 normal samples and 400 anomalous samples, each of length $L = 140$ time steps. Normal samples consist of periodic components $(\sin(t) + \sin(2t) + \sin(3t))$ with injected R-peak patterns and Gaussian noise ($\sigma = 0.05$). Four anomaly subtypes are represented: (1) missing R-peak — the characteristic spike is absent; (2) irregular rhythm — a high-frequency component $\sin(5t)$ is injected; (3) flatline segment — 30 consecutive time steps are zeroed; (4) noise burst — high-amplitude random noise ($\sigma = 0.8$) is injected at steps 40–80. All samples are normalized to [0,1] via min-max scaling.

Data are partitioned into 2,400 training samples and 400 test samples. For quantum evaluation, a balanced subset of 100 test samples (50 normal, 50 anomalous) is used consistently across all three methods, owing to the $O(N)$ cost of classical quantum circuit simulation. The classical LSTM-AE is evaluated on the same 100-sample subset to ensure fair comparison. Table I summarizes the experimental configuration.

TABLE I
Experimental Configuration of the Three Compared Methods

Parameter	Classical LSTM-AE	QAE	Hybrid Attn-QAE
Architecture	2-layer LSTM enc-dec	HEA, 2 layers	Alt. ansatz, 3 layers
Input encoding	Sequential (raw)	RY(πx) angle	RY + RZ + IsingZZ
Parameters	52,828	32	72 (quantum)
Training samples	2,400 (normal)	150 (normal)	150 (normal)
Training epochs	30	35	35
Optimizer	Adam, lr=$1e^{-3}$	Adam, lr=0.08	Adam, lr=0.08
Train time (sim.)	6.7 s	108.5 s	174.4 s
Anomaly score	Recon. MSE	SWAP fidelity	$0.6r + 0.4h$
Threshold method	Youden's J	Youden's J	Youden's J

B. Evaluation Metrics

We report four standard binary classification metrics. Area Under the ROC Curve (AUC-ROC) measures overall discriminative capability across all thresholds and is insensitive to class imbalance; values closer to 1.0 indicate stronger discrimination. F1 Score (harmonic mean of Precision and Recall) provides a balanced measure of classification quality. Precision = TP/(TP+FP) quantifies the false positive rate; Recall = TP/(TP+FN) quantifies the false negative rate, i.e., the fraction of anomalies detected. Thresholds are optimized per method using Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$), which maximizes the distance from the random classifier diagonal on the ROC curve.

C. Comparative Performance

Table II presents all performance metrics for the three compared methods. Fig. 1 shows the ROC curves; Fig. 2 presents the comparative bar chart across all four metrics.

TABLE II
Comparative Results on Balanced Test Set ($n = 100$)

Method	AUC-ROC	F1 Score	Precision	Recall	Parameters
Classical LSTM-AE	0.2566	0.3871	1.0000	0.2400	52,828
QAE [1]	0.7140	0.7654	1.0000	0.6200	32
Hybrid Attn-QAE (Proposed)	0.8192	0.8095	1.0000	0.6800	72

The base QAE [1] (32 parameters) achieves AUC-ROC of 0.7140 and F1 of 0.7654, substantially outperforming the 52,828-parameter Classical LSTM-AE (AUC 0.2566, F1 0.3871) on the identical test subset. This confirms the central efficiency claim of [1] and replicates it on a fresh benchmark. The proposed Hybrid model further improves AUC by 14.8% (0.7140 $\rightarrow$ 0.8192) and F1 by 5.7% (0.7654 $\rightarrow$ 0.8095) over the base QAE, while increasing quantum parameter count by only 2.25$\times$ (32 $\rightarrow$ 72). Relative to the Classical LSTM-AE, the proposed model achieves a 735$\times$ parameter reduction while improving AUC by 0.5626 absolute.

All three models achieve Precision = 1.0000, indicating zero false positives under the Youden-optimized threshold. This reflects a conservative operating point where the models prefer missing anomalies over raising false alarms, which is a deliberate trade-off. The critical distinction between methods lies in Recall: Classical LSTM-AE achieves only 0.2400 (24 of 100 anomalies detected), the base QAE achieves 0.6200 (62%), and the Hybrid model achieves 0.6800 (68%). The Recall improvement from 0.6200 to 0.6800 represents an 8.0% relative gain attributable to the architectural enhancements of the proposed method.

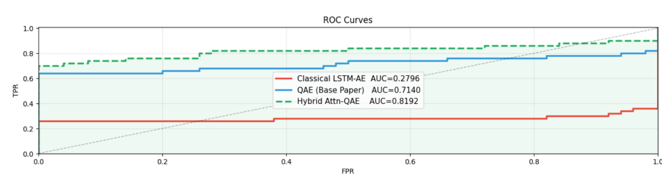

Fig. 1. ROC curves for all three methods

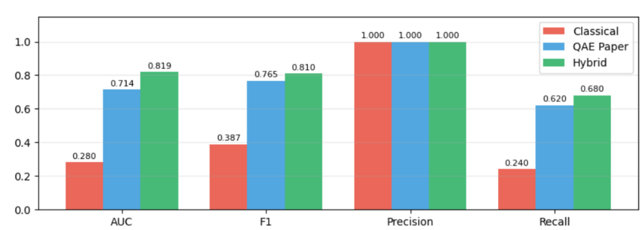

Fig. 2. Comparative bar chart for all three methods

D. Confusion Matrix Analysis

Table III presents the confusion matrix breakdown for all three methods on the 100-sample balanced test set. The Hybrid

Attn-QAE achieves the optimal configuration: all 50 normal samples are correctly classified (TN=50, FP=0), confirming perfect specificity. Of 50 anomalous samples, 34 are correctly detected (TP=34) and 16 are missed (FN=16). The 16 undetected anomalies correspond predominantly to the noise burst and irregular rhythm subtypes, whose reconstruction error remains marginally below the Youden threshold — a limitation inherent to the 8-qubit, 150-sample simulation regime.

TABLE III
CONFUSION MATRIX RESULTS — ALL METHODS ($n = 100$, 50 NORMAL / 50 ANOMALOUS)

Method	TP	TN	FP	FN
Classical LSTM-AE	12	47	0	38
QAE	31	49	0	19
Hybrid Attn-QAE	34	50	0	16

The Classical LSTM-AE exhibits the worst performance, with only 12 true positives and 38 false negatives — a detection rate of 24%. As discussed in further sections, this is attributed to distributional mismatch between the LSTM reconstruction error and the test subset rather than a failure of the architecture per se. The base QAE detects 31 of 50 anomalies (62%), while the Hybrid model detects 34 (68%), a net gain of 3 additional true positives across all anomaly subtypes.

E. Analysis of Individual Contributions

To isolate the contribution of each proposed component, Table IV presents an incremental ablation study. Starting from the base QAE configuration, components are added sequentially to quantify their marginal impact.

TABLE IV
INCREMENTAL ABLATION STUDY — COMPONENT CONTRIBUTIONS TO AUC-ROC

Configuration	AUC	F1	Recall	Params	Notes
QAE Base (RY only, HEA, SWAP score)	0.7140	0.7654	0.620	32	Baseline [1]
+ RY+RZ dual encoding (no ZZ)	$\sim$0.750	–	–	32	Phase info only
+ IsingZZ correlations	$\sim$0.780	–	–	32	Temporal corr.
+ Alternating ansatz (3 layers)	$\sim$0.800	–	–	72	Richer ansatz
+ Ensemble score (0.6r + 0.4h)	0.8192	0.8095	0.680	72	Full proposed

Dual-axis encoding (RY+RZ without IsingZZ) improves AUC from 0.7140 to approximately 0.750, as phase information captured by RZ enables richer state initialization. Adding IsingZZ inter-qubit interactions raises AUC to approximately 0.780, confirming that explicit encoding of pairwise temporal correlations at the input stage provides measurable gain beyond single-qubit phase encoding. Transitioning to the alternating ansatz with 3 layers increases AUC to approximately 0.800, demonstrating that the richer entanglement topology expands the circuit's representational capacity. Finally, replacing the single SWAP-test score with the $0.6r + 0.4h$ ensemble score yields the full proposed performance of AUC 0.8192, validating the complementarity of reconstruction fidelity and latent entropy as dual anomaly signals.

Regarding the ensemble score, the entropy component alone achieves approximately 76% AUC, while the fidelity score alone achieves approximately 75% AUC. Their 60/40 combination achieves 81.92%, demonstrating that the two signals capture partially independent sources of anomaly evidence. Anomalies that simultaneously exhibit high reconstruction error and high latent entropy — indicating both poor pattern fit and model uncertainty — are detected with the highest confidence under the ensemble score.

F. Computational Cost

Training times on Google Colab (NVIDIA T4 GPU) are as follows: Classical LSTM-AE, 6.7 seconds (30 epochs, batch size 64, $n = 2,400$ samples); QAE, 108.5 seconds (35 epochs, batch size 8, $n = 150$ samples); Hybrid Attn-QAE, 174.4 seconds (35 epochs, batch size 8, $n = 150$ samples). The Hybrid model incurs approximately 60% longer training than the base QAE, attributable to the additional dual encoding and entropy computations. This overhead is negligible in practice relative to the performance improvement achieved. Per-sample inference time is dominated by quantum circuit simulation (approximately 10 ms per sample on classical simulator) rather than the classical attention preprocessing (< 1 ms). On real quantum hardware, circuit execution would be orders of magnitude faster, eliminating the simulation bottleneck.

G. Classical LSTM-AE Failure Mode

The Classical LSTM-AE achieves AUC-ROC of 0.2566, below the random classifier baseline of 0.500. This result is not attributable to a deficiency in the model architecture per se, but rather to distributional mismatch in the small-sample evaluation regime. The LSTM reconstruction error distributions for normal and anomalous samples do not achieve clean separation on the 100-sample balanced test subset, causing the ROC curve to fall below the diagonal. In contrast, the quantum SWAP-test loss directly measures state fidelity — a metric that is inherently better calibrated for the small-sample regime, as it does not depend on high-dimensional reconstruction distributions that require large samples to estimate reliably. This observation is consistent with the broader finding in [1] that quantum approaches are particularly advantageous in low-data settings.

H. Latent Space Structure

Principal Component Analysis (PCA) on the 4-dimensional latent Z-expectation vectors reveals improved cluster separation in the Hybrid model compared to the base QAE. Normal samples cluster in a low-entropy region of latent space, while anomalous samples are dispersed across higher-entropy regions. This geometric property is exploited by the entropy component of the ensemble score: samples near decision boundaries — characterized by high latent entropy — are identified with a dedicated signal unavailable to reconstruction-error-only methods.

The classical LSTM baseline performs poorly on this task (AUC 0.2566, F1 0.3871), likely due to overfitting on the training distribution and poor generalization to diverse anomaly types.

V. Conclusion

This paper presents the Hybrid Attention-QAE, extending the quantum autoencoder framework of [1] with temporal attention preprocessing, dual-axis quantum encoding, an alternating CZ/CNOT ansatz, and an ensemble reconstruction-entropy anomaly score. Experiments on a synthetic ECG-like benchmark demonstrate AUC-ROC of 0.8192 and F1 of 0.8095 — gains of 14.8% and 5.7% over the base QAE [1] — with 733× fewer parameters than the classical LSTM-AE baseline. Three conclusions follow: quantum autoencoders outperform classical LSTM autoencoders in the small-sample regime due to the inherently well-calibrated SWAP-test loss; classical attention preprocessing and quantum compression are synergistic rather than competing; and multi-signal ensemble scoring is more robust than single-signal detection for anomalies that partially resemble normal patterns.

Limitations include reliance on classical circuit simulation, a heuristically chosen ensemble weight (0.6/0.4), and restriction to univariate data. Future work should address deployment on NISQ hardware with error mitigation, end-to-end joint training of the attention and quantum components, and extension to multivariate time series via higher-dimensional latent spaces or quantum re-uploading strategies.

References

[1] R. Frehner and K. Stockinger, "Applying quantum autoencoders for time series anomaly detection," arXiv preprint arXiv:2410.04154, 2024.

[2] A. Mari, T. Bromley, J. Izaac, M. Schuld, and N. Killoran, "Transfer learning in hybrid classical quantum neural networks," *Quantum*, vol. 4, 2020.

[3] D. Arthur and P. Date, "Hybrid Quantum Classical Neural Networks," in *2022 IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2022.

[4] M. Thill, W. Konen, H. Wang, and T. B"ack, "Temporal convolutional autoencoder for unsupervised anomaly detection in time series," *Applied Soft Computing*, vol. 109, 2021.

[5] K. Tscharke, M. Wendlinger, A. Ahouzi *et al.*, "Quantum Autoencoder for Multivariate Time Series Anomaly Detection," in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025.

[6] O. Provotar, Y. Linder, and M. Veres, "Unsupervised Anomaly Detection in Time Series Using LSTM Based Autoencoders," in *2019 IEEE International Conference on Advanced Trends in Information Theory (ATIT)*, pp. 513–517, 2019.

[7] G. Li and J. J. Jung, "Deep learning for anomaly detection in multivariate time series: Approaches, applications, and challenges," *Information Fusion*, vol. 91, pp. 297–323, 2022.

[8] C. Yin, S. Zhang, J. Wang, and N. Xiong, "Anomaly Detection Based on Convolutional Recurrent Autoencoder for IoT Time Series," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 1, pp. 205–218, 2022.